

Studying the Impact of Post-Training Quantization on Fairness in Transformer Models

Maryala Sai Teja
2024121007

Monosij Roy
2023111016

Rudra Choudhary
2024121012

Vedant Tejas
2023112018

Abstract

We propose to study how post-training quantization (PTQ) affects fairness and internal representations in Transformer-based language models. Using a pretrained BERT encoder kept frozen, and a lightweight classifier trained on top, we will apply multiple numerical-precision settings (FP32 baseline, FP16, and INT8 PTQ) and measure both classification performance and fairness-related behavior. Fairness evaluation will combine group-level metrics (demographic parity, equal opportunity), counterfactual consistency tests (prediction flip-rate under demographic swaps), and representation-level analyses (layerwise L2 / cosine shifts and Centered Kernel Alignment (CKA)). The study aims to answer whether and how PTQ alters demographic sensitivity when the encoder is not fine-tuned, and to identify where in the model representation shifts (if any) are concentrated.

1 Introduction

Transformer-based pretrained language models have become central to modern NLP systems. To deploy these models efficiently, practitioners often apply post-training quantization (PTQ) to reduce memory and compute. While many works have shown that carefully applied quantization preserves aggregate performance (e.g., accuracy or F1) for a range of tasks, less is known about how quantization affects fairness: might reducing numerical precision change how models treat demographic attributes, or alter their internal representations in ways that increase demographic disparities?

This project isolates the effect of quantization by (1) using a pretrained BERT encoder as a frozen feature extractor, and (2) varying only the numerical precision of the encoder while keeping the (lightweight) classifier training in full precision. We will evaluate not only standard task metrics but also counterfactual and representation-level measures (including CKA)

to obtain a layered view of how quantization perturbs model behavior.

Finally, to study the architecture-dependence of quantization-induced effects, we will extend our analysis beyond encoder-only BERT models to include decoder-only transformer architectures.

2 Background and Related Work

- **LLM Quantization for Code and Extreme Low-bit Regimes:** Giagnorio (2025) shows that code-generation LLMs up to 34B parameters can be quantized to 4 bits with minimal performance loss using code-specific calibration. While the study emphasizes quantization quality and calibration strategies, it does not address fairness, which is the focus of our work (Giagnorio, 2025).
- **Quantization and Fairness (Vision Models):** Liu et al. (Liu, 2025) study the fairness impact of quantization in vision models, finding that PTQ often worsens accuracy disparities across demographic groups. They further show that Quantization-Aware Training (QAT) can amplify these gaps even more than PTQ, while simple data augmentation or rebalancing strategies can partially mitigate the resulting unfairness.
- **Quantization and Fairness in Language Models:** Proskurina et al. (Proskurina et al., 2025) show that common quantization pipelines (e.g., GPTQ-style algorithms) can increase biased outputs and degrade fairness-benchmark performance for generative LLMs. They propose *Fair-GPTQ*, which incorporates explicit group-fairness constraints into the quantization objective to guide rounding and reduce stereotyped generation; their approach reduces group bias while keeping most of the efficiency benefits of low-bit quantization.

3 Objective and Motivation

We aim to determine whether post-training quantization inadvertently alters a language model’s demographic sensitivity. Specifically, we will:

- Isolate the effect of quantization by using a pretrained BERT encoder as a fixed (frozen) feature extractor and training only a lightweight classifier on top. This avoids confounding fine-tuning updates.
- Compare models under different numerical precisions (e.g. FP32 baseline, FP16, INT8 PTQ) to see how accuracy and fairness change. Prior work suggests quantization can worsen group disparities, but it’s unknown if a frozen-model setup exhibits this.
- Measure classification performance (accuracy, F1, etc.) in each case, and also evaluate fairness by subjecting test inputs to demographic perturbations (e.g. swapping names or identity terms). We will track how often the model’s prediction changes under such counterfactual swaps.
- Analyze internal representations: we will extract hidden embeddings and logits at each layer for a set of inputs and quantify how they shift between precisions. This will reveal where quantization injects noise or bias into the model’s internal state.

4 Datasets

We plan to utilize three complementary datasets to evaluate fairness at both the group and individual levels. Each dataset serves a distinct role in isolating the effects of quantization on model representations.

- **Jigsaw Unintended Bias in Toxicity Classification:** To evaluate group-level fairness metrics (Demographic Parity and Equal Opportunity), we use the Jigsaw dataset (Borkan et al., 2019). This corpus contains approximately 2 million user comments labeled for toxicity, with a subset explicitly tagged with identity labels (e.g., race, gender, religion). By treating this as a binary classification task (Toxic vs. Non-Toxic), we can measure if quantization disproportionately degrades performance for specific demographic subgroups.

- **Bias in Bios:** To evaluate the impact of quantization on stereotypical associations, we use the Bias in Bios dataset (De-Arteaga et al., 2019). This dataset consists of biographical texts annotated with one of 28 occupations (e.g., Professor, Physician, Nurse). It is widely used to study gender bias, as many occupations exhibit strong gender imbalances. We use this to measure *Equal Opportunity Difference* by comparing the True Positive Rates (TPR) for male and female instances within the same occupation before and after quantization.
- **Equity Evaluation Corpus (EEC):** To strictly measure counterfactual consistency, we employ the Equity Evaluation Corpus (Kiritchenko and Mohammad, 2018). EEC consists of 8,640 synthetic sentences generated from templates that vary only by demographic identity terms (e.g., names associated with different races or genders) while keeping the semantic context fixed. This dataset serves as a controlled test set (paired with a sentiment classifier trained on standard data like IMDB) to calculate the *Counterfactual Flip Rate* and measure representational shifts (CKA, Cosine Similarity) caused solely by numerical precision reduction.

5 Proposed Methodology

- **Model Setup:** Start with a pretrained BERT-base model (e.g. uncased). Keep all weights fixed to represent the frozen encoder. Attach a shallow classifier (e.g. one or two-layer MLP) on top of the [CLS] embedding for the target task. Train only this classifier (in full precision) on the task data.
- **Quantization:** Apply post-training quantization to the frozen BERT model. We will consider multiple schemes: full-precision FP32 (baseline), half-precision FP16, and 8-bit integers (INT8) via PTQ. For INT8 we will use a standard dynamic/static quantization toolkit (e.g. PyTorch PTQ or NVIDIA’s TensorRT) as in prior work. We do *not* fine-tune BERT after quantization.
- **Classification & Fairness Testing:** For each precision setting (FP32/FP16/INT8), run the model+classifier on the test set. Record standard performance metrics (accuracy, F1, etc.).

To evaluate fairness, for each test sentence we construct a parallel sentence with a demographic swap (e.g. changing “She” to “He”, or “John” to “Jane”). We then check if the model’s predicted label changes between the original and swapped sentences. We quantify this “flip rate” as well as group-wise prediction rates.

- **Representation Analysis:** Collect hidden representations (e.g. the [CLS] vector) from each BERT layer for a sample of inputs. Compare these vectors between FP32 and quantized models using metrics like cosine similarity and L2 distance.

For example, for layer l , we compute:

$$\Delta_l^{(2)} = \frac{1}{N} \sum_{i=1}^N \|h_l^{\text{FP32}}(x_i) - h_l^{\text{Quant}}(x_i)\|_2,$$

$$\cos_l = \frac{1}{N} \sum_{i=1}^N \frac{\langle h_l^{\text{FP32}}(x_i), h_l^{\text{Quant}}(x_i) \rangle}{\|h_l^{\text{FP32}}(x_i)\|_2 \cdot \|h_l^{\text{Quant}}(x_i)\|_2}.$$

Additionally, we will use **Centered Kernel Alignment (CKA)** to quantify the similarity between hidden representations across layers and quantization levels. CKA is invariant to orthogonal transformations and isotropic scaling, and is well-suited for comparing representational structures.

Given two representation matrices X and Y of shape $N \times d$ (for N inputs), we compute linear CKA as:

$$\text{CKA}(X, Y) = \frac{\|X^\top Y\|_F^2}{\|X^\top X\|_F \cdot \|Y^\top Y\|_F},$$

where $\|\cdot\|_F$ is the Frobenius norm. Higher CKA scores indicate greater alignment between FP32 and quantized representations. We will compute CKA per-layer and visualize alignment trends across depth and quantization settings.

- **Implementation Details:** We will follow standard PTQ procedures (calibrating with a held-out set if needed). Classifier training will be repeated with different random seeds to assess stability. Representations for analysis will be collected from a fixed sample of inputs to ensure consistency across quantization levels.

6 Evaluation

- **Performance Metrics:** Report classification accuracy and F1-score for each precision level (FP32, FP16, INT8) on held-out test data. This checks that quantization does not catastrophically degrade performance.
- **Fairness Metrics:** We measure two standard group fairness metrics:

- **Demographic Parity Difference:**

$$\left| P(\hat{Y} = 1 \mid A = 0) - P(\hat{Y} = 1 \mid A = 1) \right|$$

- **Equal Opportunity Difference:**

$$|\text{TPR}_{A=0} - \text{TPR}_{A=1}|$$

- **Counterfactual Flip Rate:** the fraction of predictions that change under a demographic token swap:

$$\frac{1}{N} \sum_{i=1}^N I(\hat{y}(x_i) \neq \hat{y}(x_i^{\text{swap}}))$$

- **Representation Shift:** For each layer l , compute the average L2 shift and cosine similarity (as defined above) between FP32 and quantized hidden states. Also compute layerwise CKA to measure representational alignment. Plot these differences to observe where quantization injects the most distortion. We expect deeper layers to accumulate more quantization noise.
- **Analysis:** We will compare how all metrics change from FP32 to FP16 to INT8. For example, do we see a sharp increase in demographic disparity at INT8 relative to FP16? We will analyze whether observed fairness degradations (if any) correlate with representational drift, group imbalance, or uncertainty.
- **Baselines and Ablations:** If fairness drops dramatically, we can consider simple mitigations (e.g. re-balancing data) to see if the effect is due to imbalance. A baseline with no sensitive attribute swaps will show aggregate bias metrics, and our counterfactual tests reveal hidden effects masked by aggregates.

References

- Daniel Borkan, Lucas Dixon, Jeffrey Sorensen, Nithum Thain, and Lucy Vasserman. 2019. Nuanced metrics for measuring unintended bias with real data for text classification. In *Companion proceedings of the 2019 world wide web conference*, pages 491–500.
- Maria De-Arteaga, Alexey Romanov, Hanna Wallach, Jennifer Chayes, Christian Borgs, Alexandra Chouldechova, Sahin Geyik, Krishnaram Kenthapadi, Adam Kalai, and Ankur Taly. 2019. Bias in bios: A case study of semantic representation bias in a high-stakes setting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, pages 120–128.
- Alessandro Giagnorio. 2025. [Extreme quantization of code-generation large language models: calibration, dataset effects, and 2–4 bit regimes](#). arXiv preprint. ArXiv:2503.07103v2.
- Svetlana Kiritchenko and Saif Mohammad. 2018. Examining gender and race bias in two hundred sentiment analysis systems. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, pages 43–53.
- Bing Liu. 2025. [On how quantization \(ptq and qat\) can exacerbate fairness disparities: theory and empirical analysis](#). In *Proceedings of ICML 2025 (Virtual)* — Poster. Poster.
- I. Proskurina and 1 others. 2025. [Fair-gptq: Incorporating group-fairness constraints into quantization objectives for large language models](#). arXiv preprint. ArXiv:2509.15206.